Chapter 3: Vector Calculus

The following subjects will be discussed

- The "del" Operator.
- Gradient, Divergence, Curl and Laplacian operators.

Examples

(1) Does the vector field $\vec{F} = xy\,\vec{a}_x - 2x\,\vec{a}_y$ exhibit any circulation behavior at point P(-2, 1, 3)?

$$\vec{\nabla}\times\vec{F} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & -2x & 0 \end{vmatrix} = -(2+x)\,\vec{a}_z \qquad \left(\vec{\nabla}\times\vec{F}\right)\Big|_P = 0$$

(No circulation behavior)

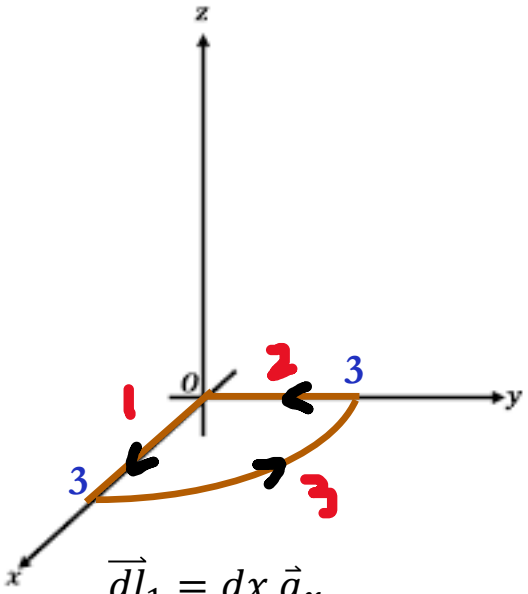
(2) Verify Stokes theorem over the given contour.

$$\int_S \left(\vec{\nabla}\times\vec{F}\right)\cdot\overrightarrow{dS} = \int_0^{\frac{\pi}{2}}\int_0^3 -(2+\rho\cos\varphi)\,\rho\,d\rho\,d\varphi = -9\left(1+\frac{\pi}{2}\right)$$

$$\oint_L \vec{F}\cdot\overrightarrow{dl} = \int_{L1}\vec{F}\cdot\overrightarrow{dl}_1 + \int_{L2}\vec{F}\cdot\overrightarrow{dl}_2 + \int_{L3}\vec{F}\cdot\overrightarrow{dl}_3$$

$$\oint_L \vec{F}\cdot\overrightarrow{dl} = \int_{L1} xy\,dx - \int_{L2} 2x\,dy + \int_{L3}(xy\,dx - 2x\,dy)$$

$$\oint_L \vec{F}\cdot\overrightarrow{dl} = \int_0^3 x\sqrt{9-x^2}\,dx - \int_3^0 2\sqrt{9-y^2}\,dy = -9\left(1+\frac{\pi}{2}\right)$$



$$\overrightarrow{dl}_1 = dx\,\vec{a}_x$$

$$\overrightarrow{dl}_2 = dy\,\vec{a}_y$$

$$\overrightarrow{dl}_3 = dx\,\vec{a}_x + dy\,\vec{a}_y$$

Vector Differentiation: Laplacian

The Laplacian is a combination of operators.

It can operate on scalar distributions. $\nabla^2 A = \vec{\nabla} \cdot \vec{\nabla} A$

In rectangular coordinated: $\nabla^2 A(x,y,z) = \frac{\partial^2}{\partial x^2} A(x,y,z) + \frac{\partial^2}{\partial y^2} A(x,y,z) + \frac{\partial^2}{\partial z^2} A(x,y,z)$

For the case $\nabla^2 A = 0$, the solution for A is sines and cosines (harmonic solution)

It can operate on vector distributions. $\vec{\nabla}^2 \vec{A} = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \vec{\nabla} \times \vec{\nabla} \times \vec{A}$

In rectangular coordinated: $\vec{\nabla}^2 \vec{A} = \nabla^2 A_x \, \vec{a}_x + \nabla^2 A_y \, \vec{a}_y + \nabla^2 A_z \, \vec{a}_z$

Example: Given the potential distribution $V(x,y,z) = e^{-z} \sin 2x \cosh y$

$$\nabla^2 V = e^{-z} \cosh y \frac{\partial^2}{\partial x^2} \sin 2x + e^{-z} \sin 2x \frac{\partial^2}{\partial y^2} \cosh y + \sin 2x \cosh y \frac{\partial^2}{\partial z^2} e^{-z}$$

$$\nabla^2 V = -2e^{-z} \sin 2x \cosh y$$

Remarks

(1) Differentiation operators are "point" operators.

(2) A vector field is uniquely specified within a given space by its divergence and curl.

(3) A vector field with $\vec{\nabla} \cdot \vec{A} = 0$, is solenoidal in nature (field lines go in closed paths).

(4) A solenoidal vector field can always be expressed as the curl of another vector function (Example: $\vec{\nabla} \cdot \vec{B} = 0 \rightarrow \vec{F} = \vec{\nabla} \times \vec{B}$).

(5) A vector field with $\vec{\nabla} \times \vec{A} = 0$, is irrotational in nature (also called conservative).

(6) A conservative vector field can always be expressed as the gradient of a scalar function (Example: $\vec{\nabla} \times \vec{E} = 0 \rightarrow \vec{E} = -\vec{\nabla} V$).

(7) A scalar function with $\nabla^2 A = 0$, is sinusoidal in nature (sine and cosine variation).

Chapter 4: Electrostatics

The following subjects will be discussed

- Coulomb's Law for Electric Force.
- Definition of Electric Field Intensity.
- Electric Field Calculations: Point Charges.
- Electric Field Calculations: Charge Distributions.
- Gauss's Law and its Applications.
- The Electric potential.

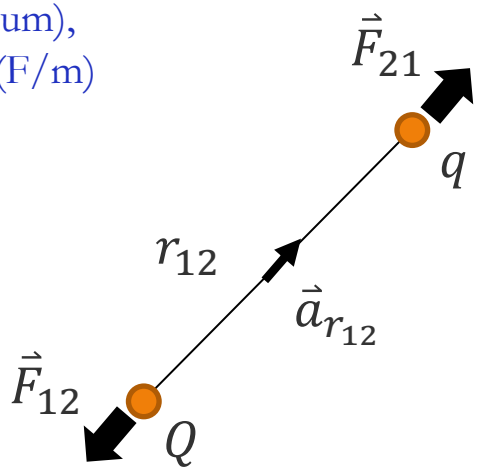
Coulomb's Law of Electrostatics

The electrostatic force between any two particle charges is given by Coulomb's Law.

$$\vec{F} = \frac{1}{4\pi\varepsilon}\frac{Q\,q}{r_{12}^2}\,\vec{a}_{r_{12}} \quad \text{(N)}$$

For free space (vacuum), $\varepsilon = \varepsilon_o = 8.85\text{x}10^{-12}$ (F/m)

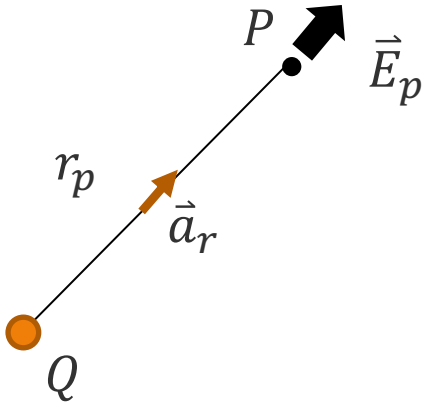
- Equal force between the charged particles
- Permittivity of the medium
- The separation distance between the particles
- A unit vector along the line connecting the particles



The electrostatic field created around a particle charge is given by

$$\vec{E}_p = \frac{1}{4\pi\varepsilon}\frac{Q}{r_p^2}\,\vec{a}_r \quad \text{(V/m)}$$

- Electric field intensity at an observation point (P)
- The separation distance between the charge and the observation point (P)



Electric Field Due to Point Charges

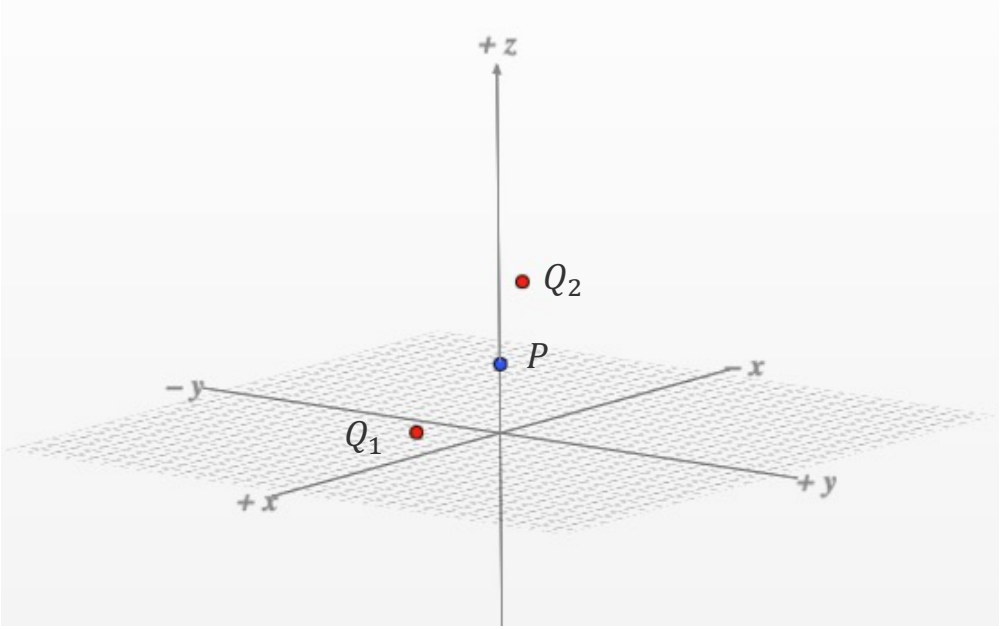
For a collection of n charge particles, the total electric field at a point P is given by

$$\vec{E}_p = \frac{1}{4\pi\varepsilon}\sum_n \frac{Q_i}{r_n^2}\vec{a}_{r_n}$$

Vector addition

Example

Calculate the total field due to the charges below at the point $P(0,0,3)$.



$Q_1 = -1\ nC$ @ $(2,-3,0)$

$Q_2 = +3\ nC$ @ $(1,2,7)$

$\vec{r}_1 = -2\,\vec{a}_x + 3\,\vec{a}_y + 3\,\vec{a}_z \rightarrow \quad r_1^2 = 22$

$\vec{r}_2 = -\vec{a}_x - 2\,\vec{a}_y - 4\,\vec{a}_z \quad \rightarrow \quad r_2^2 = 21$

$$\vec{E}_p = \frac{1}{4\pi\varepsilon}\frac{(-1\,n)(-2\,\vec{a}_x + 3\,\vec{a}_y + 3\,\vec{a}_z)}{(22)^{3/2}} + \frac{1}{4\pi\varepsilon}\frac{(3\,n)(-\vec{a}_x - 2\,\vec{a}_y - 4\,\vec{a}_z)}{(21)^{3/2}}$$

Electric Field Due to Charge Distribution: Finite Line

The differential field is given by

$$d\vec{E}_p = \frac{1}{4\pi\varepsilon}\frac{dQ}{r^2}\,\vec{a}_r = \frac{1}{4\pi\varepsilon}\frac{\rho_L\,dl}{r^2}\,\vec{a}_r \qquad dQ = \rho_L\,dl$$

$$\vec{r} = s\,\vec{a}_\rho + (z'-z)\,\vec{a}_z = s\,\vec{a}_\rho + s\tan\alpha\,\vec{a}_z$$

$$dl = dz = -s\,(\sec\alpha)^2\,d\alpha$$

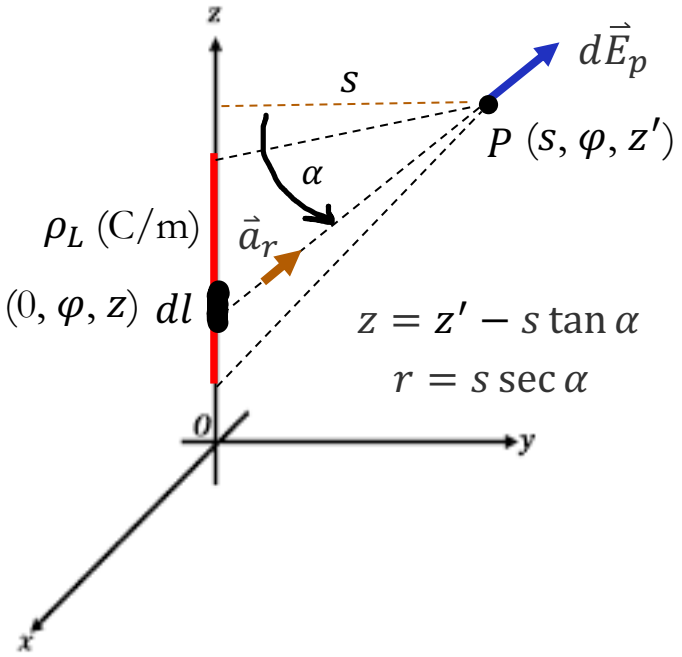
$$\vec{E}_p = \frac{1}{4\pi\varepsilon}\int_{\alpha_1}^{\alpha_2}\rho_L(-s\,(\sec\alpha)^2\,d\alpha)\frac{s\,\vec{a}_\rho + s\tan\alpha\,\vec{a}_z}{(s\sec\alpha)^3}$$

$$\vec{E}_p = \frac{-\rho_L}{4\pi\varepsilon\,s}\int_{\alpha_1}^{\alpha_2}\cos\alpha\,(\vec{a}_\rho + \tan\alpha\,\vec{a}_z)\,d\alpha$$

Assuming uniform charge density

$$\vec{E}_p = \frac{-\rho_L}{4\pi\varepsilon\,s}[(\sin\alpha_2 - \sin\alpha_1)\,\vec{a}_n - (\cos\alpha_2 - \cos\alpha_1)\,\vec{a}_L]$$

- Normal distance to axis of line
- A unit vector normal to the line
- A unit vector along the line



Special case: Infinite line.

$$\alpha_1 = 0 \qquad \alpha_2 = \pi$$

$$\vec{E}_p = \frac{\rho_L}{2\pi\varepsilon\,s}\,\vec{a}_n$$

Q & A

Send me an email if you have any questions.